

Paternal Valproate Exposure and Offspring Neurodevelopmental Outcomes

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Abstract

Background and Objectives

Evidence remains inconclusive on whether paternal valproate exposure during spermatogenesis is associated with adverse offspring outcomes, with findings lacking beyond Nordic countries. This study aimed to assess the risks of neurodevelopmental disorders (NDDs) and congenital malformations in offspring attributable to paternal exposure to valproate and other antiseizure medications (ASMs) during sperm development.

Methods

This nationwide birth cohort study used data from the Taiwan National Health Insurance Research Database, including individuals born 2001–2016 who were followed up through 2021. Paternal exposure to valproate or other ASMs was defined during the time of spermatogenesis (3 months before conception). Exposure discordant sibling sets were identified for sibling-comparison analysis to control for shared genetic and lifestyle factors. NDDs—including autism spectrum disorder (ASD), attention-deficit/hyperactivity disorder (ADHD), intellectual disability (ID), and tic disorder—and congenital malformations were defined by outpatient and inpatient medical record(s). Relative risks were estimated through Cox proportional hazards models for NDDs (hazard ratios [HRs]) and logistic regression for congenital malformations (odds ratios [ORs]).

Results

In the population cohort of 2,583,503 individuals, 1,701 were exposed to paternal use of valproate and 548 exposure-discordant sibling sets were available for sibling-comparison analysis. In population-based analyses, paternal valproate exposure was not associated with the risk of NDDs (HRs = 0.86 [95% CI 0.61–1.22] for ASD, 1.02 [0.88–1.18] for ADHD, 0.80 [0.53–1.20] for tic disorders, and 0.81 [0.55–1.19] for ID) or congenital malformations (OR = 1.07 [0.83–1.37]) in offspring. These null effects persisted when restricting analyses to children of fathers with epilepsy to control for confounding by indication, and when the sibling-comparison analysis was conducted. For other ASMs, a few associations appeared at nominal significance, but none remained in sibling-comparison analyses.

Discussion

In this large population-based study in Asia, we found no increased risk of NDDs in offspring following paternal exposure to valproate or other ASMs. These findings may contribute to ongoing debates about the management of valproate in men of reproductive age, although additional evidence is needed before drawing conclusions about changes to current practice.

Introduction

Epilepsy is one of the most common neurologic disorders affecting around 50 million people worldwide.¹ Patients with epilepsy frequently require long-term antiseizure medication (ASM)

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Glossary

ADHD = attention-deficit/hyperactivity disorder; **aHR** = adjusted hazard ratio; **aOR** = adjusted odds ratio; **ASD** = autism spectrum disorder; **ASM** = antiseizure medication; **HR** = hazard ratio; **ICD-10** = International Classification of Diseases, 10th Revision; **ID** = intellectual disability; **MHRA** = UK Medicines and Healthcare products Regulatory Agent; **NDD** = neurodevelopmental disorder; **NHIRD** = National Health Insurance Research Database; **OR** = odds ratio.

treatment through reproductive age. In the past 2 decades, the effect of teratogenicity and risk of offspring neurodevelopmental disorders (NDDs) of female patients with epilepsy in child-bearing years have been extensively studied, which resulted in significant influences on clinical practice.²⁻⁷ Nevertheless, the effect of male patients with epilepsy taking ASMs during fertile age received less attention until the UK Medicines and Healthcare products Regulatory Agent (MHRA) issued a warning against the use of valproate in new male patients younger than 55 years old.⁸ This advisory was prompted by a retrospective meta-analysis in 2023 conducted by the contract research organization, IQVIA, across Sweden, Norway, and Denmark reporting potential risk of NDD in offspring paternally exposed to valproate vs lamotrigine or levetiracetam.^{9,10} Although exemption is provided if 2 specialists independently judge that no better alternative treatment options are available or the risk does not apply, the recommendation draws different opinions.¹¹⁻¹⁷

The main arguments against strict restriction are the lack of compelling evidence due to methodology concerns (e.g., confounding by indication) and the lack of alternative effective ASMs in some patients. Population-based registry studies conducted before the warning suggested mild to no increased risk of adverse outcomes linked to paternal ASM use during conception, including valproate.¹⁸⁻²⁰ More recently, studies using the nationwide Denmark registry and systematic review found no association of valproate during spermatogenesis with offspring risk of congenital malformation or NDD, including autism.²¹⁻²³ This is reassuring but needs more studies to corroborate. Another unanswered question is whether the paternal effect of valproate is ethnicity-specific because current data are all accumulating from Nordic countries.

In this study, we used the National Health Insurance Research Database (NHIRD) of Taiwan to investigate the effect of paternal exposure to valproate and other commonly prescribed ASMs on offspring congenital malformation and NDD during spermatogenesis, using multiple study designs to address potential confounding.

Methods

Study Design

We conducted a nationwide birth cohort study among individuals born between January 1, 2001, and December 31, 2016, who were followed up until 2021, using the NHIRD, which covers approximately 99% of the residents of Taiwan.²⁴

We integrated several linked databases using a unique set of personal identifiers.

We inferred kinships using 2 databases, the Registry for Beneficiaries and Maternal and Child Health Databases, which have been validated with high accuracy >98%.²⁵ The Registry for Beneficiaries contain information on the identifiers of the relationships between insured person and their dependents; only spouses and blood relatives are eligible as dependents of an insured person. The Maternal and Child Health Databases contain information on the unique personal identifier of infants and both the biological parents. We selected families wherein the parent and child had an age difference of at least 12 years to maximize the probability to obtain correct familial relatedness. Full-siblings were defined as 2 individuals having the same parents.

We retrieved the insurance amount and urbanization level of the residents using the Registry for Beneficiaries database. Detailed health conditions of the newborns, along with characteristics of pregnancy and labor, including birth weight, parity, gestational age, delivery method, pregnancy and delivery complications, and Apgar score, were retrieved using the Birth Certificate Application database. Disease diagnoses were identified using the inpatient and outpatient claims databases. We used data from death certificates to monitor the vital status of the study samples.

We used several complementary study designs to account for confounding and evaluated the association of paternal ASM use and NDD risks in offspring: (1) population-based analysis with adjustment for measured confounders, (2) population analysis restricted to fathers with epilepsy diagnoses to further account for confounding by indication, and (3) sibling-comparison analysis to account for residual or unmeasured familial confounders. Fathers with epilepsy were identified as having at least 2 outpatient or 1 inpatient diagnostic codes (ICD-9 code: 345; ICD-10 code: G40) in the NHIRD. In the sibling design, we compared offspring exposed to paternal ASM use during spermatogenesis with their unexposed siblings to assess NDD outcomes, assuming that the exposure-discordant sibling sets shared similar genetic backgrounds and familial environments.

To further explore the presence of unmeasured confounding factors for paternal ASM use during the time of spermatogenesis, we examined the association of paternal ASM use during the mother's pregnancy with offspring NDD risk

(i.e., comparing children whose father did vs did not use ASMs during pregnancy) as a sensitivity analysis.²⁶ Analyses were conducted with or without adjustment of paternal ASM use during the spermatogenesis window, in both the population-based and sibling-comparison settings. Any observed nonnull associations would indicate residual confounding rather than a causal spermatogenic effect.

Standard Protocol Approvals, Registrations, and Patient Consents

This study was approved by the Research Ethics Committee, National Health Research Institutes, Taiwan (EC1120607-E). The requirement for informed consent was waived because the NHIRD contains deidentified data.

Study Samples

Eligible individuals born 2001–2016 with parental information and maternal characteristics of pregnancy and labor were identified in the NHIRD. We excluded (1) twins, triplets, or high-order multiples; (2) infants with chromosomal abnormality; (3) still births; (4) infants with gestational age <32 weeks or >42 weeks; (5) infants with birth weight <1,500 g; and (6) mothers who were foreigners. All study samples were followed up to the diagnosis of study outcome, death, or until the end of the year 2021.

Paternal ASM Use

Any paternal ASM use during the time of spermatogenesis (i.e., 3 months before conception, back calculated from the date of birth, and gestational weeks) was ascertained regardless of whether ASMs were used alone or in combination. Specific ASM drugs were considered, including valproate, phenytoin, carbamazepine, phenobarbital, lamotrigine, oxcarbazepine, topiramate, clobazam, vigabatrin, gabapentin, levetiracetam, and pregabalin (eTable 1). For valproate, we additionally identified paternal use restricted to monotherapy during spermatogenesis.

Any paternal ASM use during the time of maternal pregnancy of the child was examined as a sensitivity analysis. We assumed that paternal ASM use during the time of maternal pregnancy would have no association with NDD risks in children.

Study Outcome

Incident NDD cases, including autism spectrum disorders (ASDs; ICD-9 code: 299; ICD-10 code: F84), attention-deficit/hyperactivity disorders (ADHDs; ICD-9 code: 314; ICD-10 code: F90), tic disorders (ICD-9 code: 307.2; ICD-10 code: F95), and intellectual disability (ID; ICD-9 code: 317–319; ICD-10 code: F70–F79), were defined by the presence of at least 2 outpatient or 1 inpatient medical record(s). Age at first diagnosis was recorded for each NDD and analyzed as a time-to-event outcome. Congenital malformation (diagnosis during the first 90 days after delivery, detailed ICD codes in eTable 2)²⁷ was considered as a binary outcome.

Covariates

Family history of epilepsy (ICD-9 code: 345; ICD-10 code: G40) and major psychiatric disorders, including schizophrenia

(SCZ; ICD-9 code: 295; ICD-10 code: F20 and F25), bipolar disorder (BPD; ICD-9 code: 296.0–296.1 and 296.4–296.8; ICD-10 code: F30, F31, and F34.0), ASD, and ADHD was considered. The lookback period for family histories ranged from 1998 to 2021.

Other covariates included offspring sex, birth year (2001–2004, 2005–2008, 2009–2012, and 2013–2016), paternal age (categorized as < 20, 20–24, 25–29, 30–34, 35–39, 40–45, and ≥45), maternal age (categorized as < 20, 20–24, 25–29, 30–34, 35–39, and ≥40), maternal lifestyles during pregnancy (alcohol use, drug misuse, and tobacco use), birth season, parity (categorized as 1, 2, and ≥3), delivery methods (classified as unassisted vaginal birth, assisted vaginal birth, and cesarian section), 1-minute Apgar score (<7 and ≥7), gestational age (categorized as 33–36, 37–40, and >40 weeks), birth weight, pregnancy complications (yes and no), delivery complications (yes and no), urbanization level (classified as urban, suburban, and rural), and insurance level (categorized as tertiles).

Statistical Analysis

The distribution of demographics, pregnancy, labor, and birth characteristics, as well as parental history of epilepsy and psychiatric disorders, was stratified by paternal ASM use and described by numbers and percentages. Univariate analysis was performed to assess feature comparisons between paternal-ASM exposed and unexposed groups using chi-squared tests and Fisher exact tests.

In the population-based analysis, we estimated the adjusted hazard ratio (aHR) with a 95% CI for paternal ASM use about offspring's hazard for each NDD outcome, including ASD, ADHD, tic, and ID, using Cox proportional hazard models with chronologic age as the timescale. Censoring was defined at death or the end of follow-up, whichever comes first. For the binary outcome, congenital malformation, we estimated the adjusted odds ratio (aOR) and the associated 95% CI using logistic regression models. Robust standard errors were applied to account for familial clustering. Adjustments were made for offspring sex, birth year, paternal and maternal age, maternal smoking, alcoholism, and drug addiction during pregnancy, birth season, parity, delivery method, 1-minute APGAR score, gestational age, birthweight, pregnancy and delivery complications, urbanization level of the residential area, family's insurance amount, and parental and maternal history of epilepsy and psychiatric disorders in the regression models.

In the sibling-comparison analysis, we performed stratified Cox regression models within each set of full siblings from the same family for ASD, ADHD, tic, and ID and performed conditional logistic regression models for congenital malformation. This compared the NDD risk in individuals exposed to paternal ASM use during spermatogenesis with their siblings without exposure to paternal ASM use. Adjustments were made for offspring sex, birth year,

maternal smoking, alcoholism, and drug addiction during pregnancy, birth season, parity, delivery method, 1-minute APGAR score, gestational age, birthweight, and pregnancy and delivery complications.

For the sensitivity analysis, we estimated the effect of paternal ASM use during the mother's pregnancy, with or without adjusting for paternal ASM use during the spermatogenesis window, on top of the same covariates in either the population-based or sib-comparison models.

The nominal significance level was set at 0.05. To account for multiple testing of the 5 NDD outcomes, a Bonferroni-corrected threshold of 0.01 (0.05/5) was applied. All statistical analyses were performed using the SAS statistical package (version 9.4; SAS Institute Inc., Cary, NC).

Data Availability

Individual deidentified data from the NHIRD cannot be shared publicly due to legal restrictions. The NHIRD used in this study is held by the Taiwan Ministry of Health and Welfare and under controlled access. Researchers interested in accessing the data set can submit an application form to the Ministry of Health and Welfare requesting access.

Results

Study Cohort Characteristics

We identified a total of 2,903,757 eligible individuals born 2001–2016 with linked paternal and maternal information in the NHIRD. After study-specific exclusion criteria (Methods), 2,583,503 live-born singletons were included as the population cohort across 1,721,170 families, among whom 1,621,603 had at least 1 sibling across 761,535 families (eFigure 1). Among all 2,583,503 offspring, 6,987 (0.27%) were exposed to any paternal ASM use during the spermatogenesis window (i.e., the 3 months before conception and back calculated from the date of birth) and 1,701 (0.07%) were specifically exposed to paternal use of valproate. Valproate was among the most commonly prescribed ASMs, along with carbamazepine and phenytoin, while vigabatrin and pregabalin were the least frequently used (eTable 3). The median follow-up for the exposed group was 12.78 (± 4.75) years. The remaining 2,576,516 offspring comprised the ASM-unexposed group, with a median follow-up of 12.36 (± 4.61) years.

Compared with the unexposed group, children whose father were prescribed ASMs were linked to younger (<20 years of age) or more advanced (>40) paternal/maternal ages, a higher proportion of rural residence, lower family insurance premiums, and higher rates of maternal pregnancy tobacco use and pregnancy complications. The exposed group was also associated with a significantly higher prevalence of paternal epilepsy and psychiatric history, and to a lesser extent, maternal diagnoses (Table 1; eTable 4).

Paternal ASM Use and Risk of Neurodevelopmental Disorders

We identified 41,979 children (1.62%) diagnosed with ASD, 197,717 (7.65%) with ADHD, 38,956 (1.51%) with tic disorders, and 28,698 (1.11%) with ID in the full cohort during the study period (eTable 5). A total of 31,984 children (1.24%) were born to fathers with a prior diagnosis of epilepsy.

In population-based analyses, paternal use of valproate or any other ASM during spermatogenesis—whether examined individually or in aggregate—was not associated with the risk of any of the 4 NDDs in offspring after adjustment for potential confounders (p values >0.05 ; Figure 1A; eTable 6). For valproate specifically, the aHRs were 0.86 (95% CI 0.61–1.22) for ASD, 1.02 (0.88–1.18) for ADHD, 0.80 (0.53–1.20) for tic disorders, and 0.81 (0.55–1.19) for ID. Population analyses restricted to offspring with epilepsy-affected fathers similarly showed no association between ASM exposure and NDD outcomes. Notably, had parental psychiatric history not been included as a covariate, significant associations emerged, with paternal use of any ASM or valproate linked to an increased ADHD risk in offspring (aHR = 1.23 [95% CI 1.06–1.43]; eTable 7).

Sibling Comparison Analysis: Neurodevelopmental Disorders

For the sibling-comparison analysis, 1,621,603 offspring across 761,535 families had at least 1 sibling in the full cohort. Among these, 6,021 children were identified across 2,716 exposure-discordant families (i.e., sibling sets and the effective sample size) in which a father had different ASM exposure status between siblings during spermatogenesis, including 1,220 children across 548 exposure-discordant families specifically for valproate (eFigure 1; eTable 8).

When the comparison was conducted within these sibling sets—where exposed children were compared with their unexposed siblings—the associations observed in population study disappeared, yielding consistent null findings across ASMs (e.g., for valproate: aHRs were 0.75 [0.26–2.13] for ASD, 1.01 [0.65–1.58] for ADHD, 0.64 [0.20–2.09] for tic disorders, and 0.55 [0.21–1.43] for ID; Figure 1A; eTable 6).

When limiting to valproate monotherapy, a significant association with ADHD was observed in the population analysis (aHR = 1.28 [1.08–1.52], p value = 0.004); however, this signal did not hold in the restricted population cohort (aHR = 1.04 [0.75–1.44], p value = 0.832) or sibling comparison analyses (aHR = 0.98 [0.61–1.60], p value = 0.947), suggesting it is likely a result of residual confounding (eTable 6).

Sensitivity Analysis: Neurodevelopmental Disorders

For the sensitivity analysis, we identified 13,651 offspring (0.52% of 2,583,503) in the population cohort whose fathers were prescribed ASM during the mother's pregnancy (i.e., an

Table 1 Characteristics of Offspring Exposed or Unexposed to Paternal Use of Antiseizure Medications (ASMs) During Spermatogenesis

	Number of offspring identified in the full cohort (%)		
Characteristics	Unexposed to paternal ASM use (n = 2,576,516)	Exposed to paternal ASM use (n = 6,987)	p Value ^a
Sex			
Male	1,342,281 (52.10)	3,639 (52.08)	0.9809
Female	1,234,235 (47.90)	3,348 (47.92)	
Birth year			
2001–2004	600,399 (23.30)	1,925 (27.55)	<0.0001
2005–2008	648,620 (25.17)	1,702 (24.36)	
2009–2012	637,190 (24.73)	1,634 (23.39)	
2013–2016	690,307 (26.79)	1,726 (24.70)	
Paternal age (y)			
<20	10,875 (0.42)	31 (0.44)	<0.0001
20–24	148,106 (5.75)	490 (7.01)	
25–29	606,087 (23.52)	1,486 (21.27)	
30–34	1,000,583 (38.83)	2,479 (35.48)	
35–39	597,271 (23.18)	1,699 (24.32)	
40–44	171,270 (6.65)	602 (8.62)	
≥45	42,324 (1.64)	200 (2.86)	
Maternal ages (y)			
<20	46,644 (1.81)	138 (1.98)	<0.0001
20–24	328,548 (12.75)	1,035 (14.81)	
25–29	845,108 (32.80)	2,193 (31.39)	
30–34	945,411 (36.69)	2,399 (34.34)	
35–39	360,556 (13.99)	1,035 (14.81)	
≥40	50,249 (1.95)	187 (2.68)	
Urbanization level of residential area			
Urban	1,421,240 (55.16)	3,653 (52.28)	<0.0001
Suburban	949,912 (36.87)	2,603 (37.25)	
Rural	205,364 (7.97)	731 (10.46)	
Family's insurance premium (NTD)			
<26,400	676,098 (26.24)	2,173 (31.10)	<0.0001
26,400–45,800	925,987 (35.94)	2,664 (38.13)	
>45,800	974,431 (37.82)	2,150 (30.77)	
Parental history of epilepsy			
No	2,547,359 (98.87)	4,160 (59.54)	<0.0001
Yes	29,157 (1.13)	2,827 (40.46)	
Maternal history of epilepsy			
No	2,548,143 (98.90)	6,874 (98.38)	<0.0001

Continued

Table 1 Characteristics of Offspring Exposed or Unexposed to Paternal Use of Antiseizure Medications (ASMs) During Spermatogenesis (continued)

Characteristics	Number of offspring identified in the full cohort (%)		p Value ^a
	Unexposed to paternal ASM use (n = 2,576,516)	Exposed to paternal ASM use (n = 6,987)	
Yes	28,373 (1.10)	113 (1.62)	
Parental psychiatric history			
SCZ/BPD/ASD/ADHD	32,193 (1.25)	999 (14.30)	<0.0001
Maternal psychiatric history			
SCZ/BPD/ASD/ADHD	44,728 (1.74)	215 (3.08)	<0.0001

Abbreviation: ADHD = attention-deficit/hyperactivity disorders; ASD = autism spectrum disorder; BPD = bipolar disorder; SCZ = schizophrenia.
^ap Values were calculated using chi-squared tests or Fisher exact tests when 1 or more expected cell frequencies were less than 5.

exposure incapable of causing the outcome yet influenced by the same biases that may affect the original association; eFigure 1). Compared with the 2,569,942 children of fathers who did not use ASMs, these offspring showed characteristics similar to those observed with paternal ASM use during sperm formation. In addition, they were more likely to be associated with maternal drug misuse during pregnancy, lower APGAR scores, and shorter or longer gestational weeks (eTable 9).

Consistent with the primary population-based analysis, paternal valproate use during pregnancy was not associated with NDD risk in offspring (aHRs: 0.89 [0.68–1.17] for ASD, 1.03 [0.91–1.16] for ADHD, 1.06 [0.80–1.41] for tic disorders, and 0.99 [0.75–1.31] for ID), with only minimal changes after additional adjustment for paternal ASM use during spermatogenesis (Figure 1B; eTable 10). A few ASM associations appeared at nominal significance, most notably between gabapentin use and tic disorders (aHR = 1.56 [1.09–2.21], *p* value = 0.014; with additional adjustment: aHR = 1.59 [1.12–2.27], *p* value = 0.010), although these did not withstand multiple testing correction.

For sibling comparison, 13,166 children were identified across 5,929 exposure-discordant families for paternal ASM use during pregnancy, including 2,295 children across exposure-discordant 1,020 families specifically for valproate (eTable 11). When examined among these discordant sibling sets, the sensitivity analysis likewise revealed no significant associations (eTable 12), indicating robustness of the primary sibling-comparison results.

Paternal ASM Use and Risk of Congenital Malformations

A total of 95,562 children (3.70%) were diagnosed with congenital malformations within 90 days after birth in the full cohort during the study period. In population-based analyses, paternal use of valproate during spermatogenesis was not significantly associated with congenital malformation risk in offspring (aOR = 1.07 [0.83–1.37]). The null findings persisted when analyses were restricted to children born of fathers with

epilepsy (aOR = 1.32 [0.94–1.84]) or when assessed within exposure-discordant sibling sets (aOR = 1.29 [0.66–2.49]; Figure 2A; eTable 7). For other ASMs, nominally significant associations were observed for any ASM use (aOR 1.24 [1.03–1.50], *p* value = 0.024) and phenytoin use (aOR 1.39 [1.04–1.85], *p* value = 0.025) among offspring of fathers with epilepsy, but neither survived Bonferroni correction. Sibling-comparison analyses showed null results across ASM types.

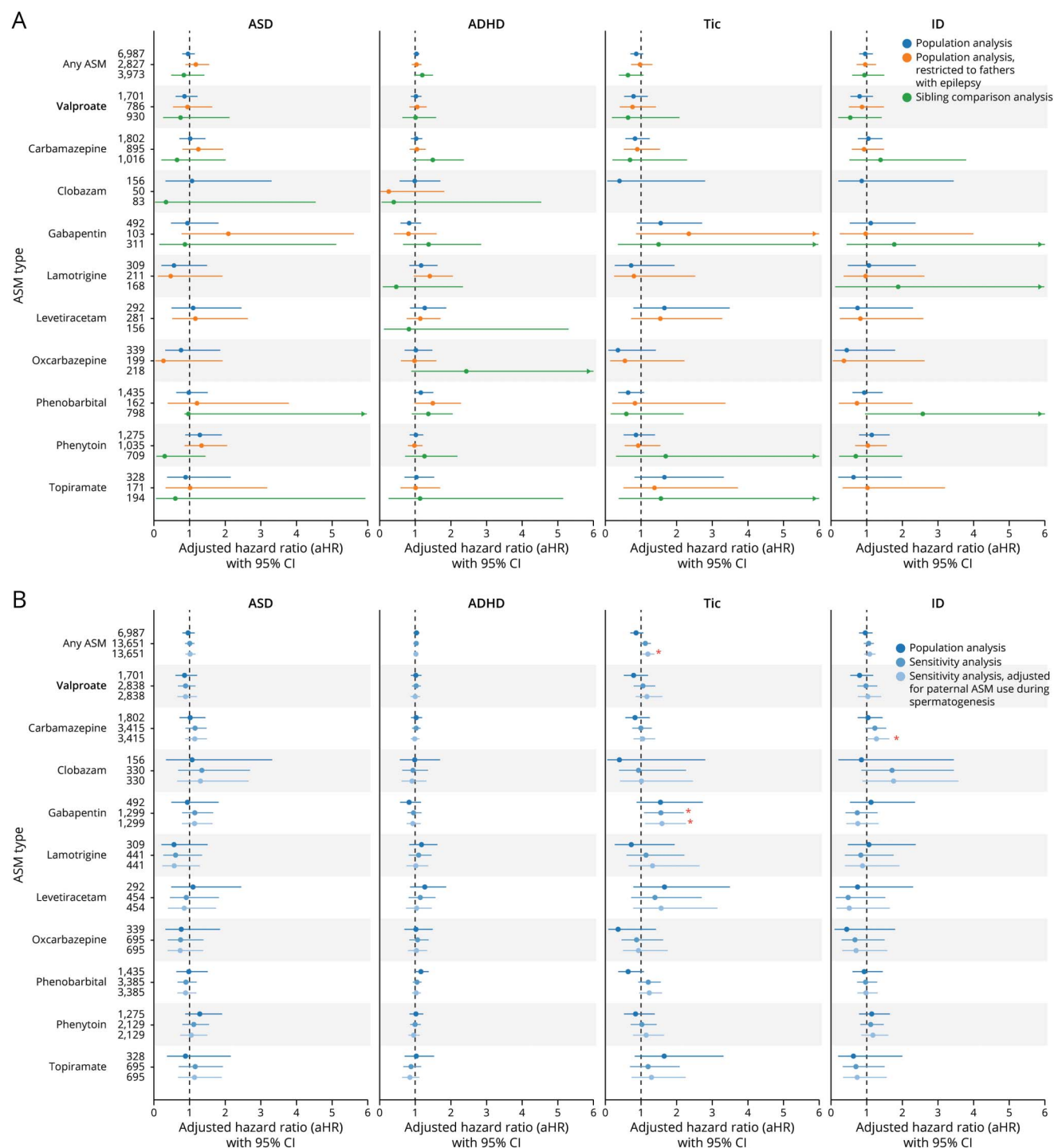
Sensitivity Analysis: Congenital Malformations

In population-based sensitivity analyses, paternal valproate use during the mother’s pregnancy was not associated with an increased risk of congenital malformation in offspring (aORs = 1.02 [0.83–1.24] and 1.00 [0.81–1.24] before and after adjustment of paternal use during spermatogenesis), nor was use of any other ASM (Figure 2B; eTable 10). Similarly, sibling-based sensitivity analyses showed no association across ASM types with congenital malformations (eTable 12), suggesting limited influence of unmeasured confounding in the primary analysis.

Discussion

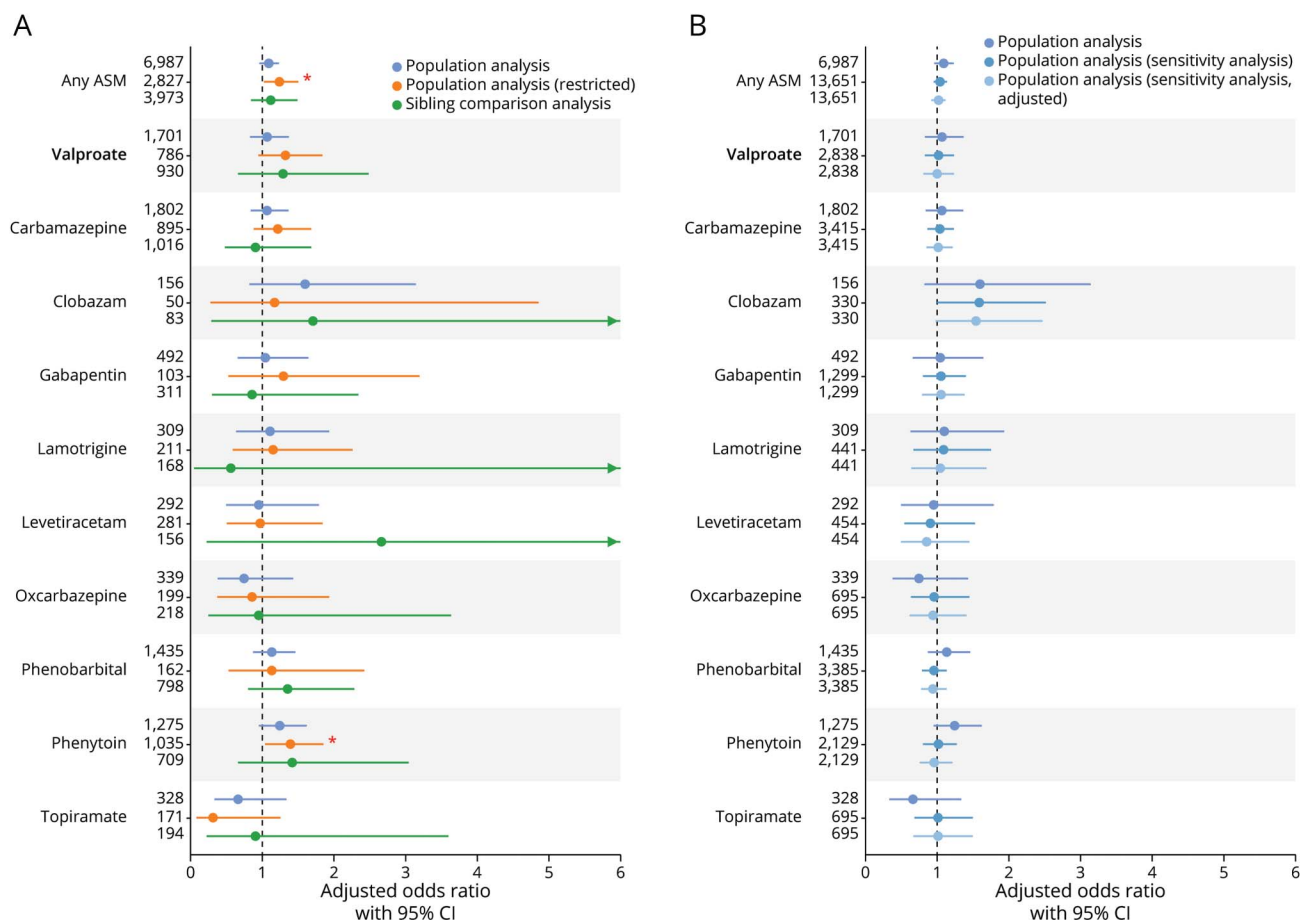
A recent warning from the UK MRHA advised against the use of valproate in men aged 55 and younger over concerns of impaired fertility and adverse outcomes in offspring. This announcement has generated debates due to insufficient evidence and underscored the need for data especially beyond European-based studies. In this population-based study with more than 2.5 million births in Taiwan (2001–2016) identified from the nationwide health insurance registry, we found no increase in risk of NDDs or congenital malformations among offspring born of fathers prescribed valproate or other commonly used ASMs during the time of spermatogenesis. These findings were consistent across population-based analyses, including restriction to fathers with epilepsy to address confounding by indication, and sibling-comparison analyses to account for genetic and familial factors—with robustness suggested by sensitivity analysis evaluations in both population and sibling-based settings.

Figure 1 Effects of Paternal ASM Use on Neurodevelopmental Risk in Offspring



(A) Primary analysis: adjusted hazard ratios (aHRs) and their associated 95% CIs for paternal ASM use during spermatogenesis on offspring risk of NDDs were illustrated across 3 designs, including (1) population-based analysis, (2) population-based analysis restricted to fathers with an epilepsy diagnosis, and (3) sibling-comparison analysis. All 3 designs were properly adjusted for relevant confounding factors (see Methods). The full list of statistics and sample sizes is available in eTable 6. (B) Sensitivity analysis: aHRs and their associated 95% CIs were shown for (1) the primary population-based analysis, along with 2 sensitivity analyses in the population cohort: (2) the effects of paternal ASM use during the mother's pregnancy, adjusting for the same covariates as in the primary analysis, and (3) the effects of paternal ASM use during the mother's pregnancy, additionally adjusted for paternal ASM use during spermatogenesis. The full list of statistics and sample sizes is available in eTable 10. Only nonmissing and nonzero estimates were displayed on the plot. Effects of pregabalin and vigabatrin were largely missing because of low counts and were not depicted. Values displayed on the y-axis represent the number of offspring exposed to each paternal ASM use during spermatogenesis under each study design. Upper bounds of the 95% CIs exceeding the value of 6 were indicated by a rightward arrow. ADHD = attention-deficit/hyperactivity disorder; ASD = autism spectrum disorder; ASM = antiepileptic medication; ID = intellectual disability; NDDs = neurodevelopmental disorders.

Figure 2 Effects of Paternal ASM Use on the Risk of Congenital Malformations in Offspring



(A) Primary analysis: adjusted hazard ratios (aHRs) and their associated 95% CIs for paternal ASM use during spermatogenesis on offspring risk of congenital malformations were illustrated across 3 designs, including (1) population-based analysis, (2) population-based analysis restricted to fathers with an epilepsy diagnosis, and (3) sibling-comparison analysis. All 3 designs were properly adjusted for relevant confounding factors (see Methods). The full list of statistics and sample sizes is available in eTable 6. (B) Sensitivity analysis: aHRs and their associated 95% CIs were shown for (1) the primary population-based analysis, along with 2 sensitivity analyses in the population cohort: (2) the effects of paternal ASM use during the mother's pregnancy, adjusting for the same covariates as in the primary analysis, and (3) the effects of paternal ASM use during the mother's pregnancy, additionally adjusted for paternal ASM use during spermatogenesis. The full list of statistics and sample sizes is available in eTable 10. Only nonmissing and nonzero estimates were displayed on the plot. The effects of pregabalin and vigabatrin were largely missing because of low counts and were not depicted. Values displayed on the y-axis represent the number of offspring exposed to each paternal ASM use during spermatogenesis under each study design. Upper bounds of the 95% CIs exceeding the value of 6 were indicated by a rightward arrow. Significant effects at a nominal p value <0.05 were marked by an asterisk sign (*). None of the effects surpassed the Bonferroni-corrected threshold of 0.01. ASM = antiepilepsy medication.

In the past 2 decades, research from international pregnancy registries in patients with epilepsy has successfully changed the prescription behavior of ASMs and improved outcomes in offspring born to women with epilepsy.^{6,7,28,29} Recent regulatory policy changes in the United Kingdom have drawn attention to the use of valproate in fathers with epilepsy at the time of conception. Our study is the first large-scale Asian population cohort to explore this important issue, finding no association of paternal valproate use during spermatogenesis with offspring NDD outcomes (e.g., for ASD: aHR = 0.86 [95% CI 0.61–1.22] in our study vs 0.92 [95% CI 0.65–1.30] in the Danish study). Notably, our findings contrast with the initial meta-analysis report by IQVIA across Denmark, Norway, and Sweden⁹ (now published in a peer-reviewed journal¹⁰) that prompted the MHRA warning, where higher risk for any NDD (aHR = 1.50 [95% CI 1.09–2.07]) was observed

in offspring paternally exposed to valproate vs lamotrigine or levetiracetam. These discordant findings might be due to various methodologic differences, such as exposure and outcome definitions, confounding and indication bias adjustment, and study design and statistical modeling. On the other hand, our results align with recent population studies from Denmark and Sweden, where null effects of paternal valproate use on offspring NDD risk were reported during spermatogenesis or at conception^{20,21} and persisted even after following the IQVIA protocol.²² Similarly, a recent systematic review found no clear evidence that paternal ASM use poses a significant risk to offspring outcomes.²³

This result is reassuring to epileptologists because valproate remains an effective ASM for many epilepsy syndromes with onset in childhood or adolescence, such as genetic generalized

epilepsy and juvenile myoclonic epilepsy. If the risk were substantial, physicians would need to avoid valproate or switch ASMs before family planning, but this may lead to less optimal seizure control. Currently, there is no registry that specifically follows young male patients with epilepsy to prospectively assess major congenital malformations or neurodevelopmental consequences in their children. Therefore, evidence from national registries across different populations provides important guides for neurologists to counsel patients and prevent unnecessary harm and anxiety.

Beyond valproate, we also did not identify other ASMs that significantly increase the risk of congenital malformations or NDDs in offspring of male patients with epilepsy. Analyses for certain ASMs, however, may be underpowered due to a small number of patients, and further studies are warranted. Overall, compared with maternal exposure, paternal exposure to ASMs seems to have a lesser impact on adverse offspring outcomes, possibly due to the shorter exposure window during spermatogenesis and the absence of direct *in utero* exposure to ASMs during fetal development.

Our study has several strengths. We conducted a nationwide birth cohort study using Taiwan's single-payer compulsory insurance program with an exceptionally high coverage rate, thus limiting selection bias. Several potential confounding factors were considered, including perinatal factors, family socioeconomic status, and parental disease history. In addition, complementary designs were applied to evaluate the role of residual and unmeasured familial factors in the etiology of NDDs.

Nevertheless, some limitations should be noted. First, the study participants were born between 2001 and 2016 and were followed until 2021. The limited follow-up duration, with a maximum of 21 years and a minimum of 5 years, may not be sufficient to capture all NDD diagnoses. Future studies with longer follow-up durations are needed. Second, formal neuropsychological test results were not available because of the diagnostic-code-based nature of our cohort study, and mild neurodevelopmental deficits may therefore go undetected. Third, we did not examine the potential dose-dependent effect of ASMs nor did we adjust the family history of intellectual disability in relation to offspring outcomes. Fourth, our sibling analyses may be subject to carryover effects.³⁰ Families already having an older child with NDDs or congenital malformations may pay more attention to preparing for and during subsequent pregnancies, as well as to their parenting behaviors. Finally, time-varying and non-shared familial confounding factors were not considered in our sibling analyses.

In conclusion, our findings did not support an increased risk of paternal exposure to valproate—or other ASMs—being associated with a higher likelihood of NDD diagnosis in the offspring. The selection of ASMs for young male patients with epilepsy should not be dominated by concerns over

potential future fatherhood unless more robust evidence emerges. We believe that these pieces of evidence call for a reconsideration of the regulation on valproate use in male patients with reproductive potential by relevant authorities. Future population-scale studies across diverse nations and ethnicities may strengthen or refute the association, which will improve our care for all patients with epilepsy of reproductive age.

Author Contributions

Y.-C.A. Feng: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. M.-C. Lin: analysis or interpretation of data. C.-H. Tseng: analysis or interpretation of data. C.-S. Wu: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. M.-H. Tsai: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. S.-H. Wang: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data.

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Disclosure

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